

# Mathematical proof - Set 1 - Solutions

Specification Prf1-Prf4.

## Answer key

| Question | Answer | Topic              |
|----------|--------|--------------------|
| 1        | E      | Mathematical proof |
| 2        | D      | Mathematical proof |
| 3        | A      | Mathematical proof |
| 4        | E      | Mathematical proof |
| 5        | D      | Mathematical proof |
| 6        | D      | Mathematical proof |
| 7        | B      | Mathematical proof |
| 8        | B      | Mathematical proof |
| 9        | A      | Mathematical proof |
| 10       | B      | Mathematical proof |
| 11       | A      | Mathematical proof |
| 12       | A      | Mathematical proof |

## Worked solutions

### Question 1 PRF-01 · Mathematical proof · spec Prf1

Consider the claim: **for every positive integer  $n$ ,  $n^2 + n$  is even.** Which of the following is a complete and valid proof?

**Answer: E**

A covers every  $n$  with a general argument. B checks only three cases; C handles only even  $n$ ; D is false for odd  $n$ ; E is incomplete.

### Question 2 PRF-02 · Mathematical proof · spec Prf1

Which of the following is a counterexample to the claim **if  $n$  is a positive integer, then  $2^n + 1$  is prime**?

**Answer: D**

$2^3 + 1 = 9 = 3 \times 3$  is not prime. ( $n = 1, 2, 4, 8$  give 3, 5, 17, 257, all prime.)

### Question 3 PRF-03 · Mathematical proof · spec Prf1

Here is a proof that  $\sqrt{2}$  is irrational, with its steps in the wrong order.

- I Then  $p^2 = 2q^2$ , so  $p^2$  is even, so  $p$  is even; write  $p = 2k$ .
- II Suppose, for contradiction, that  $\sqrt{2} = \frac{p}{q}$  where  $p$  and  $q$  are positive integers with no common factor.
- III This contradicts the assumption that  $p$  and  $q$  have no common factor, so  $\sqrt{2}$  is irrational.
- IV Then  $4k^2 = 2q^2$ , so  $q^2 = 2k^2$  is even, so  $q$  is even.

Which is the correct order?

**Answer: A**

Assume (II), deduce  $p$  even (I), deduce  $q$  even (IV), reach the contradiction (III).

### Question 4 PRF-04 · Mathematical proof · spec Prf1

A student wants to prove that **the square of any odd integer leaves remainder 1 when divided by 8**. Which of the following is the most appropriate first step?

**Answer: E**

A direct proof:  $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$ , and  $k(k + 1)$  is even, so the square is  $8m + 1$ . Checking cases is not a proof; assuming the claim is circular.

### Question 5 PRF-07 · Mathematical proof · spec Prf1

Prove by cases that  $n^3 - n$  is divisible by 3 for every integer  $n$ . Which set of cases is both sufficient and necessary to consider?

**Answer: D**

$n^3 - n = (n - 1)n(n + 1)$  is a product of three consecutive integers; splitting by remainder on division by 3 shows one factor is a multiple of 3 in every case.

**Question 6** PRF-08 · Mathematical proof · spec Prf2

It is given that for all real  $x$ : if  $f(x) > 0$  then  $g(x) > 0$ . It is also known that  $g(2) = -1$ . What can be deduced?

**Answer: D**

Contrapositive: if  $g(x) \leq 0$  then  $f(x) \leq 0$ . Since  $g(2) = -1 \leq 0$ ,  $f(2) \leq 0$  (it could be 0 or negative).

**Question 7** PRF-13 · Mathematical proof · spec Prf1

Which of the following is a correct proof that **the sum of any three consecutive integers is divisible by 3**?

**Answer: B**

A is general and correct. B checks cases; C uses non-consecutive integers; D's reasoning is invalid (containing a multiple of 3 does not make the sum a multiple of 3).

**Question 8** PRF-14 · Mathematical proof · spec Prf1

To prove **'if  $n^2$  is even then  $n$  is even'** by contradiction, which assumption should be made?

**Answer: B**

Assume the hypothesis holds and the conclusion fails, then derive a contradiction:  $n$  odd gives  $n^2$  odd, contradicting  $n^2$  even.

**Question 9** PRF-16 · Mathematical proof · spec Prf4

A proof that **'if  $a$  and  $b$  are positive real numbers then  $\frac{a+b}{2} \geq \sqrt{ab}$ '** is given with its lines scrambled.

- I** Hence  $a + b \geq 2\sqrt{ab}$ , and dividing by 2 gives the result.
- II** Expanding,  $a - 2\sqrt{ab} + b \geq 0$ .
- III** Since squares are non-negative,  $(\sqrt{a} - \sqrt{b})^2 \geq 0$ .

Which order gives a valid proof?

**Answer: A**

Start from a known truth (III), expand (II), rearrange (I). Starting from the result and working backwards (I, II, III) would need every step to be reversible and is presented in the wrong direction.

**Question 10** PRF-17 · Mathematical proof · spec Prf1

Which of the following claims is FALSE, and therefore can be disproved by a counterexample?

**Answer: B**

$a = 1, b = -2$ :  $1 > -2$  but  $1 < 4$ . The others are true.

**Question 11** PRF-20 · Mathematical proof · spec Prf3

$T_n$  denotes the  $n$ th triangular number  $\frac{n(n+1)}{2}$ . A student observes  $T_1 + T_2 = 4$ ,  $T_2 + T_3 = 9$ ,  $T_3 + T_4 = 16$  and conjectures that  $T_n + T_{n+1}$  is always a perfect square. Which of the following proves the conjecture?

**Answer: A**

Only A is a general algebraic verification.

**Question 12** X2-27 · Mathematical proof · spec Prf1

Consider the claim: \*\*if  $n$  is a positive integer then  $n^2 + n + 1$  is odd.\*\* Which of the following is a valid proof?

**Answer: A**

A is a direct general argument. B only checks cases; C is irrelevant; D is correct for odd  $n$  but says nothing about even  $n$ ; E does not reach a contradiction.